Our Environment

The big idea

An **ecosystem** is all the living things (**biotic**) in an area plus the non-living surroundings (**abiotic**: air, water, soil, temperature). Energy flows through it in one direction, and our actions can harm it.

Food chains and webs

A **food chain** shows who eats whom: grass → deer → lion. Each step is a **trophic level**.

- **Producers** (plants) make food using sunlight.
- **Consumers**: herbivores (primary), carnivores (secondary, tertiary), omnivores.
- **Decomposers** (bacteria, fungi) break down dead matter and return nutrients to the soil.

Real ecosystems have **food webs**: many interlinked chains.

THE 10% LAW

Only about **10%** of the energy at one trophic level passes on to the next; the rest is used for life processes or lost as heat. That's why food chains rarely have more than **3–4 levels**, and why there are fewer animals at higher levels.

ENERGY THROUGH A CHAIN

If plants trap 10,000 J, herbivores get about 1,000 J, small carnivores 100 J and large carnivores only 10 J.

Biological magnification

Some harmful chemicals (like pesticides) are **not biodegradable**. They enter the food chain and **accumulate**, becoming more concentrated at each trophic level. Humans, at the top of many chains, carry the highest concentration.

The ozone layer

Ozone (O_3) high in the atmosphere **shields Earth from the Sun's ultraviolet (UV) radiation**, which can cause skin cancer. **Chlorofluorocarbons (CFCs)** from refrigerants and fire extinguishers damage it. In 1987, an international agreement (under UNEP) aimed to freeze CFC production at 1986 levels.

Managing waste

- **Biodegradable** waste (vegetable peels, paper, cotton) is broken down by microbes.
- **Non-biodegradable** waste (plastics, DDT, glass) persists and harms the environment.

Use the three Rs: **reduce, reuse, recycle**. Segregate waste, use cloth bags, compost kitchen waste.

Quick recap

- Ecosystem = biotic + abiotic components.
- Energy flows one way; only about 10% passes to the next level.
- Non-biodegradable chemicals magnify up the food chain.
- CFCs deplete the ozone layer, which protects us from UV.

Practice questions

1. Which of these are decomposers?
   (a) Green plants (b) Bacteria and fungi (c) Herbivores (d) Carnivores

2. If a producer traps 20,000 J of energy, a secondary consumer gets about
   (a) 2,000 J (b) 200 J (c) 20 J (d) 20,000 J

3. Biological magnification means harmful chemicals
   (a) decrease along a food chain (b) become more concentrated at higher trophic levels (c) are broken down by decomposers (d) stay in the soil only

4. The ozone layer is being damaged mainly by
   (a) carbon dioxide (b) chlorofluorocarbons (CFCs) (c) oxygen (d) water vapour

5. Which waste is non-biodegradable?
   (a) Fruit peels (b) Paper (c) Plastic bags (d) Cotton cloth

Answers: 1. b 2. b 3. b 4. b 5. c